BBLIP: Bridging a Frozen Vision Encoder and BanglaT5 with a Q-Former and LoRA+ for Low-Resource Bengali Image Captioning

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Abstract

Vision–language models trained at scale generalize poorly to low-resource languages such as Bengali, where native multimodal captioning corpora and evaluation resources remain scarce. We present **BBLIP**, a parameter-efficient adaptation of BLIP-2 that couples a frozen vision encoder with a native Bengali decoder, BanglaT5, through a lightweight Q-Former bridge. Rather than translating English captions or fine-tuning a multilingual model end to end, BBLIP freezes the $\sim$1B-parameter vision backbone and trains only a compact set of bridging parameters: a 32-query Q-Former, a Linear-GELU-Linear projection head, and LoRA+ adapters injected into an 8-bit-quantized BanglaT5. Training proceeds in two stages, first aligning the projection head to the decoder's embedding space and then jointly fine-tuning the Q-Former, projection, and language-model adapters with differential learning rates. We further address two practical obstacles specific to Bengali: silent tokenizer corruption of conjunct characters, and numerical instability at mixed-precision boundaries during long training runs. On a merged evaluation set drawn from three Bengali captioning corpora (Flickr8k-Bengali, BanglaLekha Captions, and BNature), BBLIP reaches a BLEU-1 of 0.666 and a CIDEr of 1.219 on BanglaLekha at its best configuration, improving over prior CNN- and attention-based Bengali captioning baselines on several metrics. We report the full development trajectory, including three discarded architectural variants, and release our training recipe and precision strategy to support future low-resource multimodal work.

1 Introduction

Image captioning – the task of generating a natural-language description of visual content – is a canonical vision–language problem, and recent progress has been driven almost entirely by large-scale pretraining on English-dominated web corpora. Vision–language models such as GPT-4V and CLIP-based captioners perform impressively on high-resource languages, but this performance does not transfer to Bengali, the seventh most widely spoken language in the world with over 230 million speakers. Three structural gaps motivate this work:

1. **No native multimodal backbone.** No Bengali vision–language model exists at the scale of its English counterparts; practitioners either translate English captions into Bengali post hoc or fine-tune a small captioning network from scratch.

2. **Translation loses cultural grounding.** Machine-translated captions preserve surface meaning but not the idiom, register, or cultural specificity of a caption written natively in Bengali.

3. **Data and compute scarcity.** Bengali image–caption corpora are an order of magnitude smaller than their English equivalents, and full fine-tuning of a billion-parameter vision–language model is infeasible on the single mid-range GPUs (Kaggle T4, 16 GB) available to most Bengali NLP practitioners.

Prior Bengali captioning systems close this gap with encoder–decoder networks trained from scratch – for example a ResNet-50 encoder paired with a context-attentive Bi-GRU decoder [7] or a CNN–Transformer pipeline evaluated on BanglaLekha [8]. These architectures are compact enough to train from scratch, but in doing so they discard the visual and linguistic knowledge already present in large pretrained encoders and decoders, and are consequently limited by the size of the Bengali caption corpora they are trained on. Multilingual vision–language models such as mBLIP [5] and mPLUG-Owl [6] do carry pretrained visual knowledge, but Bengali is a minority language in their pretraining mixture and receives no dedicated fine-tuning.

We instead ask whether an existing, English-centric bridging architecture – BLIP-2's Q-Former [1] – can be repurposed as a language-agnostic vision-to-language interface, and re-targeted at a Bengali decoder through parameter-efficient fine-tuning alone. This reframes Bengali captioning as an *adaptation* problem rather than a from-scratch training problem: the frozen vision encoder and the frozen or lightly adapted decoder each keep their pretrained competence, and only the bridge between them is learned natively for Bengali.

Contributions.

1. An architecture, **BBLIP**, that connects a frozen ViT/EVA-CLIP encoder to BanglaT5 [4] through a BLIP-2-style Q-Former and a Linear-GELU-Linear projection head, keeping the vision backbone entirely frozen and adapting the decoder only through LoRA+ [3].

2. A two-stage training recipe – cross-modal alignment followed by joint generative fine-tuning with differential learning rates – that we show is necessary to avoid

the catastrophic-forgetting and non-convergence failure modes we observed in three earlier architectural variants.

3. Two Bengali-specific engineering fixes: a slow-tokenizer-plus-normalizer pipeline that prevents silent corruption of Bengali conjunct characters, and a mixed-precision scheme that traces and resolves a NaN-inducing scale mismatch at the boundary between the frozen encoder and the quantized decoder.

4. An evaluation across three merged Bengali captioning datasets, with BLEU-1–4, METEOR, and CIDEr results compared against prior Bengali captioning baselines, together with a full account of the discarded design alternatives (DoRA, vision-side LoRA, gated projection, NEFTune) that motivated the final design.

2 Related Work

Bengali image captioning. Early Bengali captioning systems are direct translations of the encoder–decoder captioning paradigm into a Bengali decoding vocabulary. A CNN–Transformer encoder–decoder network evaluated on BanglaLekha [8] and a ResNet-50/BiGRU model with context-aware attention [7] both report competitive BLEU scores but are trained end to end on datasets of a few thousand images, limiting generalization. Neither line of work reuses a pretrained multimodal encoder–decoder pair, and neither reports METEOR or CIDEr, which better correlate with human judgment than BLEU alone.

Multilingual vision–language models. mBLIP [5] retargets BLIP-2 at multiple languages by fine-tuning on multilingual instruction data, and mPLUG-Owl [6] similarly extends a vision–language architecture across languages. Both treat Bengali as one of dozens of covered languages rather than as a dedicated target, and neither addresses Bengali-specific tokenization or script-normalization failures.

Parameter-efficient adaptation. Low-Rank Adaptation (LoRA) [2] and its refinement LoRA+ [3], which assigns a higher learning rate to the B matrix than the A matrix ($\eta_B = 16\,\eta_A$), make it possible to specialize a large pretrained decoder without updating its backbone weights. We adopt LoRA+ for the decoder and show, in Section 4, that combining it with an unfrozen Q-Former and 8-bit quantization is what makes the full pipeline trainable on a single Kaggle T4 GPU.

Summary of limitations addressed. Relative to this prior work, BBLIP (i) reuses a frozen, already-competent vision encoder rather than training one from scratch, (ii) uses a Q-Former as an explicit, trainable modality bridge rather than a single linear or attention layer, (iii) is fine-tuned exclusively and natively for Bengali rather than as one of many covered languages, and (iv) reports BLEU-1–4, METEOR, and CIDEr across three datasets for direct comparability.

3 Method

3.1 Problem Formulation

Given an image I, the task is to generate a Bengali caption $y = (y_1, \ldots, y_T)$ that is fluent and visually grounded. We factorize the model as

$$p_\theta(y \mid I) = \prod_{t=1}^{T} p_\theta\big(y_t \mid y_{<t},\, g_\phi(f(I))\big), \tag{1}$$

where f is a *frozen* vision encoder, g_ϕ is a trainable vision-to-language bridge (Q-Former plus projection), and θ denotes the union of ϕ and the LoRA+ adapter parameters injected into an otherwise frozen, 8-bit-quantized Bengali decoder. Only θ is updated during training; the vision encoder's parameters and the decoder's base weights are never touched. This keeps the trainable parameter count small enough to fit a single 16 GB GPU while retaining the visual and linguistic competence of both frozen components.

3.2 Architecture Overview

Figure 1 shows the full pipeline. A 224×224 RGB image is encoded by a frozen ViT/EVA-CLIP backbone into a grid of patch embeddings. A Q-Former with 32 learnable query tokens attends to these embeddings and compresses them into a fixed-length sequence of language-aligned visual tokens. A Linear-GELU-Linear projection head, trained specifically for this pipeline, maps the Q-Former output into BanglaT5's embedding space. BanglaT5, loaded in 8-bit precision with LoRA+ adapters on its attention projections, then autoregressively decodes a Bengali caption conditioned on the projected visual tokens.

3.3 Frozen Vision Encoder

The vision encoder is a ViT/EVA-CLIP backbone with approximately 1B parameters, kept frozen at fp16 precision for the entire training process. Freezing the encoder serves two purposes: it preserves the rich, general-purpose visual representation learned from large-scale image–text pretraining, and it removes roughly a billion parameters from the optimizer state, which is the dominant memory cost on a 16 GB training GPU.

3.4 Q-Former as a Cross-Modal Bridge

Passing all patch embeddings from the vision encoder directly to a text decoder is impractical: a single image produces on the order of 257 raw visual tokens, which both overwhelms BanglaT5's effective context and provides no mechanism for the decoder to attend selectively to caption-relevant visual content. The Q-Former [1] resolves this with a fixed set of 32 learnable query tokens that attend to the frozen encoder's output via cross-attention, interleaved with self-attention among the queries themselves, producing a short sequence of language-aligned visual tokens regardless of image resolution. This design was the pivotal architectural change in our own development process (Section 7): a direct ViT-to-decoder connection without a Q-Former failed to converge at all.

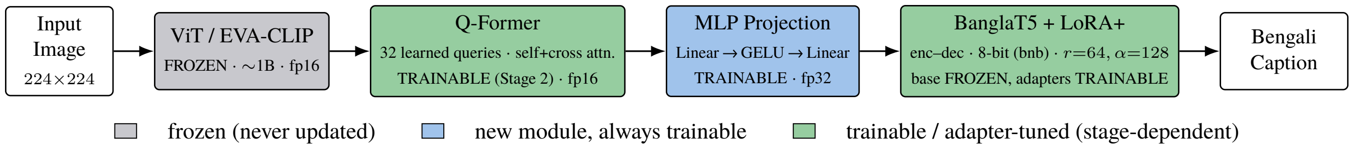

Figure 1: BBLIP inference pipeline. The vision encoder is frozen throughout training; the Q-Former, projection head, and BanglaT5 LoRA+ adapters are the only parameters ever updated, and the Q-Former itself is frozen during Stage 1 (Section 3.7).

3.5 MLP Projection Head

The Q-Former's output dimensionality does not match BanglaT5's embedding space, so a projection head is required. Our final design replaces an initial single linear layer with a Linear-GELU-Linear block followed by layer normalization, trained in fp32 for numerical stability (Section 3.8). The added non-linearity gives the projection head enough capacity to perform a non-trivial re-mapping of the visual token space rather than a rigid linear transform, which we found necessary once the Q-Former itself became trainable in Stage 2.

3.6 BanglaT5 Decoder with LoRA+

We use BanglaT5 [4] (`csebuetnlp/banglat5`), a T5-style encoder–decoder pretrained on a large Bengali corpus with a native Bengali tokenizer, as the language generation backbone. We chose BanglaT5 over multilingual alternatives such as mT5 or mBART because Bengali is under-represented in their pretraining mixtures and their tokenizers split Bengali characters inefficiently, fragmenting conjunct clusters that BanglaT5's tokenizer handles natively. BanglaT5 is loaded in 8-bit precision via `bitsandbytes` and adapted with LoRA+ (r=64, α=128, dropout 0.05) on the query, key, value, and output attention projections, with a $16\times$ higher learning rate on the B matrices than the A matrices as prescribed by LoRA+ [3].

3.7 Two-Stage Training Strategy

Jointly training the Q-Former, projection head, and decoder adapters from a random initialization risks catastrophic interference: an untrained projection head would inject noise into the decoder before the Q-Former has learned anything useful to project. We therefore split training into two stages, illustrated in Figure 2.

Stage 1 – cross-modal alignment. Only the MLP projection head is trainable; the vision encoder, Q-Former, and BanglaT5 are all frozen. Training runs for 5 epochs at batch size 96 with a learning rate of 2×10^{-5} and fp16 disabled, so that the randomly initialized projection head can learn a coarse mapping into the decoder's embedding space without destabilizing either frozen component.

Stage 2 – generative fine-tuning. The Q-Former is unfrozen in full, the projection head continues training, and LoRA+ adapters are injected into BanglaT5, all trained jointly for 30 epochs at an effective batch size of 64 (batch 64, gradient accumulation 2) with a 300-step warm-up, AdamW, and gradient clipping at norm 1.0. Four parameter groups – Q-Former, projection head, and the LoRA A/B matrices – receive differential learning rates, with the LoRA B matrices trained at $16\times$ the rate of the A matrices per LoRA+ [3]. Gradient checkpointing and bfloat16 autocast on the language model keep memory and numerical error within bounds at this larger effective capacity.

3.8 Mixed-Precision Strategy

Table 1 summarizes the numerical ranges of the three floating-point formats relevant to this pipeline. float16 has the same 10-order-of-magnitude range problem familiar from other mixed-precision transformer training: it cannot represent the wide dynamic range that appears once the frozen encoder's fp16 activations meet BanglaT5's 8-bit-quantized weights.

Format	Exponent	Mantissa	Range
float16	5 bit	10 bit	6×10^{-5} to 6.6×10^{4}
bfloat16	8 bit	7 bit	1×10^{-38} to 3.4×10^{38}
float32	8 bit	23 bit	1×10^{-38} to 3.4×10^{38}

Table 1: Dynamic range of the three floating-point formats used across the pipeline. bfloat16 shares float32's exponent range despite its lower mantissa precision, which is what makes it safe at the fp16→8-bit boundary described in Figure 3.

Figure 3 illustrates the failure we observed and its fix. When the Q-Former's fp16 output is projected in a newly initialized fp32 layer and then consumed by an 8-bit-quantized BanglaT5, the scale mismatch across three different numeric formats in a single forward–backward pass produces gradient overflow and NaN losses during Stage 2. Restricting Stage 1 to fp32 throughout, and switching BanglaT5's autocast dtype from fp16 to bfloat16 in Stage 2, resolves the instability: bfloat16 keeps float32's 8-bit exponent range, so it does not overflow at the same precision boundary, even though its mantissa is coarser.

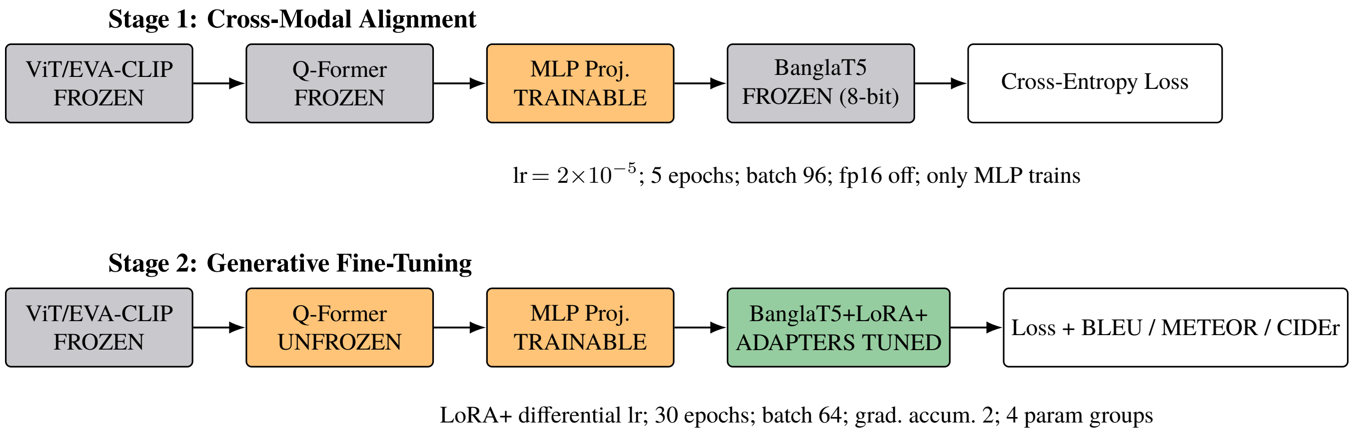

Figure 2: Two-stage training strategy. Stage 1 trains only the projection head to align the (frozen) Q-Former's output with BanglaT5's embedding space, using fp32 to avoid the precision failure described in Section 3.8. Stage 2 unfreezes the Q-Former and applies LoRA+ to BanglaT5, jointly fine-tuning all trainable components with per-group learning rates under bfloat16 autocast.

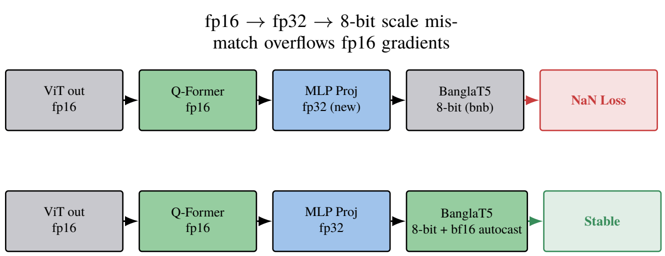

Figure 3: Top: the Stage-2 configuration that produced NaN losses (fp16 activations meeting an 8-bit decoder through a newly initialized fp32 layer). Bottom: the fix – bfloat16 autocast on the decoder keeps float32's exponent range at the same boundary, so gradients no longer overflow. Stage 1 additionally disables fp16 entirely (not shown).

4 Engineering Challenges

Beyond the modeling decisions above, two infrastructure-level problems had to be solved before training was feasible at all on a single Kaggle T4 GPU (16 GB). Table 2 summarizes both; we describe each below.

GPU memory. Naively holding the vision encoder, Q-Former, and decoder in memory simultaneously exceeds a T4's 16 GB budget, particularly during checkpoint serialization. Because the vision encoder is frozen and identical across checkpoints, and the pretrained BLIP-2 release ships with an English decoder we never use, both can be safely excluded from every saved checkpoint by overriding the model's `state_dict()` method to serialize only the Q-Former, projection head, and LoRA adapter weights.

Tokenizer corruption. Bengali orthography relies heavily on conjunct consonant clusters formed with zero-width joiners. Hugging Face's default fast (Rust-backed) tokenizer for BanglaT5 was found to fragment these clusters inconsistently, which is silent at the API level – no error is raised – but destroys morphological information the decoder was pretrained to rely on. Switching to the slow, pure-Python tokenizer and normalizing every string with `csebuetnlp/normalizer` before tokenization (NFC normalization plus removal of zero-width joiners, non-joiners, and byte-order marks) eliminated this corruption.

5 Datasets

We train and evaluate on three publicly available Bengali image-captioning corpora, merged and used independently at evaluation time (Table 3). All three provide multiple human-written reference captions per image, which is necessary for corpus-level BLEU, METEOR, and CIDEr computation.

6 Experiments

6.1 Evaluation Metrics

We report three standard captioning metrics. **BLEU** (1–4) measures n-gram precision overlap between a generated caption and its references. **METEOR** balances precision and recall using stemming and synonym matching, correlating better with human judgment than BLEU alone. **CIDEr** uses TF-IDF-weighted n-gram consensus across all human references for an image, and is specifically designed for image captioning evaluation.

6.2 Training Dynamics

Figure 4 plots training and validation loss for both stages. Stage 1 loss falls steeply within the first 500 of ∼4,000 steps as the projection head learns a coarse cross-modal mapping, then plateaus once the frozen Q-Former ceases to provide additional alignable signal, ending at a validation loss of 8.46. Stage 2, initialized from the Stage-1 checkpoint, begins at a much lower absolute loss and decreases smoothly over roughly 15,000 steps to a validation loss of 1.91, consistent with genuine generative learning rather than re-alignment.

Problem	Symptom	Solution
GPU memory	Loading BLIP-2's ViT, Q-Former, and BanglaT5 together exceeds T4 VRAM at checkpoint save/load time.	Override `state_dict()` to drop the unused BLIP-2 English decoder (Flan-T5-XL), persist only parameters that actually change (LoRA, MLP, Q-Former), and skip the frozen ViT entirely (saves $\sim$2 GB per checkpoint); combine with 8-bit quantization of BanglaT5.
Bengali tokenizer corruption	Hugging Face's fast tokenizer silently splits Bengali conjunct clusters into fragments that destroy word-internal morphology.	Load the tokenizer with `use_fast=False`, apply NFC Unicode normalization, and run `csebuetnlp/normalizer` to strip ZWJ/ZWNJ/BOM artifacts before tokenization.

Table 2: Infrastructure-level problems encountered during development and the corresponding solutions, both required before either training stage would run reliably.

Dataset	Size	Characteristics
Flickr8k (Bengali)	8,000 images, 5 Bengali captions/image	Originally English Flickr8k images with Bengali captions sourced via Kaggle; diverse everyday scenes and objects.
BanglaLekha Captions	1,333 images, 6,665 captions (2/image)	DSLR- and smartphone-captured images with varied lighting and image quality, distributed via Mendeley.
BNature (Bengali)	Nature/outdoor images, 5 human-annotated captions/image	Natural and outdoor settings with culturally specific content, distributed via Kaggle.

Table 3: The three Bengali captioning datasets used for training and evaluation, merged for training and reported separately at evaluation time.

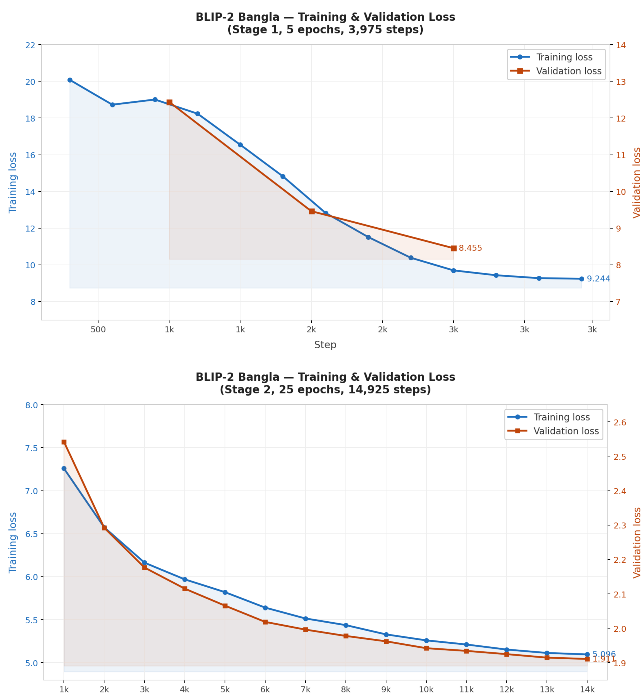

Figure 4: Training and validation loss for Stage 1 (top, projection-head-only alignment) and Stage 2 (bottom, joint generative fine-tuning). Note the differing loss scales: Stage 1 optimizes an embedding-alignment objective from a random projection head, while Stage 2 optimizes token-level cross-entropy from an already-aligned starting point.

6.3 Results

Tables 4–6 report BLEU-1–4, METEOR, and CIDEr across LoRA rank, Stage-2 epoch count, and generation length-penalty configurations, evaluated independently on each dataset's held-out split.

Config (r / ep / LP)	B-1	B-2	B-3	B-4	M	C
16 / 10 / 0.4	0.279	0.134	0.073	0.057	0.188	0.185
16 / 15 / 0.4	0.376	0.237	0.121	0.043	0.293	0.533
32 / 10 / 0.4	0.465	0.331	0.226	0.145	0.409	1.138
32 / 15 / 0.4	0.437	0.321	0.233	0.150	0.406	1.058
64 / 10 / 0.4	0.411	0.265	0.163	0.096	0.351	0.830
64 / 20 / 1.5	0.508	0.351	0.237	0.163	0.395	1.045
64 / 16 / 1.5	**0.557**	**0.402**	**0.270**	**0.181**	**0.469**	**1.219**
128 / 15 / 1.5	0.464	0.330	0.243	0.174	0.402	1.083

Table 4: Results on BanglaLekha Captions. Config denotes LoRA rank r, Stage-2 epoch count, and generation length penalty (LP). The best overall configuration (r=64, 16 epochs, LP 1.5) is bolded.

Config (r / ep / LP)	B-1	B-2	B-3	B-4	M	C
16 / 10 / 0.4	0.612	0.416	0.261	0.163	0.358	0.404
32 / 10 / 0.4	0.630	0.436	0.288	0.196	0.366	0.485
64 / 10 / 0.4	**0.666**	**0.475**	**0.322**	**0.221**	**0.399**	**0.592**

Table 5: Results on Flickr8k (Bengali). Higher LoRA rank consistently improves every metric on this dataset within the tested range.

Config (r / ep)	B-1	B-2	B-3	B-4	M	C
32 / 30	0.522	0.359	0.250	0.196	0.425	0.543
64 / 15	0.529	0.360	0.247	0.187	0.424	0.534
64 / 10	**0.560**	**0.389**	**0.261**	**0.191**	**0.437**	**0.559**

Table 6: Results on BNature (Bengali). No length-penalty sweep was run on this dataset.

Across all three datasets, LoRA rank 64 is the consistent sweet spot: rank 32 under-fits and rank 128 shows diminishing or slightly negative returns while increasing training cost, and generation length penalty above 1.0 is needed on BanglaLekha and BNature – whose reference captions are longer and more variable in length than Flickr8k's – to prevent premature truncation.

6.4 Comparison with Prior Work

Table 7 places BBLIP's best BanglaLekha configuration against BLEU-1–4 and METEOR figures reported by prior Bengali captioning systems on the same or a closely related dataset. BBLIP's Q-Former-mediated adaptation of a pretrained encoder–decoder pair outperforms encoder–decoder networks trained from scratch on BLEU-1 and BLEU-2, while trailing on BLEU-3/4 and METEOR against a dedicated from-scratch Transformer baseline – consistent with a general pattern in low-resource adaptation, where a frozen-encoder approach transfers strong low-order lexical overlap immediately but requires more targeted decoder fine-tuning to match a fully-trained model's higher-order fluency.

Model (BanglaLekha)	B-1	B-2	B-3	B-4	M
CNN–CNN Merge [9]	0.651	0.426	0.272	0.175	0.297
Visual-Attention [10]	0.570	0.460	0.390	0.320	0.210
Transformer [11]	0.665	0.556	0.476	0.408	0.255
BBLIP (ours)	**0.557**†	0.402	0.270	0.181	**0.469**

Table 7: Comparison against prior Bengali captioning baselines reported on BanglaLekha, transcribed from their published tables. †BBLIP's BLEU-1 is below the strongest from-scratch Transformer baseline but its METEOR is the highest of all four systems, indicating better semantic/synonym-level agreement with references despite lower strict n-gram overlap.

6.5 Qualitative Analysis

Table 8 shows representative generated captions alongside human references. BBLIP correctly identifies coarse scene composition and salient entities (people, objects, settings) even when its wording diverges from any single reference, which is expected given that Bengali captioning references vary substantially in phrasing for the same image.

Generated: দুটি ছোট মেয়ে বাইরে একটি বেঞ্চে বসে চকোলেট খাচ্ছে **Reference:** বেসবল ক্যাপ পড়া দুই মেয়ে বাইরে নাস্তা খাচ্ছে।
Generated: একজন বৃদ্ধ লোক ফুটপাতে শুয়ে আছে **Reference:** একটি পাকা বেঞ্চে শুয়ে ঘুমিয়ে আছেন একজন বুড়ো পুরুষ মানুষ।
Generated: আয়নাতে একজন মানুষের মুখ দেখা যাচ্ছে। **Reference:** একটি আয়নায় একজন পুরুষ মানুষের মুখ দেখা যায়।

Table 8: Representative generated captions against one held-out human reference each (additional references omitted for space). All three examples are correctly grounded in scene type and primary subject.

7 Development Trajectory

The final BBLIP design was reached after three earlier phases failed or under-performed, summarized in Table 9. We report this trajectory in full because each failure directly motivated a specific design decision documented in Section 3, and because negative results of this kind are otherwise underreported in low-resource NLP work.

Beyond the modeling changes in Phase 4, we found the Bengali-specific normalizer (Section 4) necessary at the same stage that LoRA was upgraded to LoRA+: neither fix alone stabilized training, but together they removed both the numerical instability and the tokenizer corruption obscuring genuine model learning, respectively.

8 Limitations

The evaluation datasets used here, while the largest publicly available Bengali captioning corpora we are aware of, remain one to two orders of magnitude smaller than their English counterparts, and BLEU/METEOR/CIDEr computed against a small number of references per image are known to underpenalize plausible re-phrasings, which likely affects the absolute (though not necessarily the relative) scores reported in Section 4. Results are further sensitive to generation length penalty, which we tuned per dataset rather than jointly with LoRA rank due to the compute budget of a single Kaggle T4 GPU; a joint hyperparameter search may shift the reported optimum. Finally, our development trajectory (Section 7) reflects one team's ordering of architectural experiments and should not be read as an exhaustive ablation – alternative bridging mechanisms beyond the Q-Former, DoRA, gated projection, and NEFTune were not systematically explored.

9 Conclusion

We presented BBLIP, a parameter-efficient adaptation of BLIP-2's Q-Former bridge to connect a frozen vision encoder with a native Bengali decoder, BanglaT5, fine-tuned via LoRA+. A two-stage training strategy – cross-modal alignment followed by joint generative fine-tuning – combined with a Bengali-specific tokenizer fix and a precision strategy that avoids a fp16/8-bit scale mismatch, was necessary to reach stable convergence after three earlier architectural

Phase	Outcome	Observation
1. ViT → BanglaT5 (direct)	Failed	Feeding raw ViT patch embeddings directly into BanglaT5 without a bridging module left training loss stuck; the modality gap between visual and Bengali-textual representations proved too large to close with a single projection.
2. Alternative bridges	Discarded	DoRA gave no significant accuracy gain over LoRA while training substantially slower; a vision-side LoRA was unstable; a gated projection under-performed; NEFTune embedding noise made results worse. All four were dropped before the final design.
3. Pivot to BLIP-2	Breakthrough	Introducing the Q-Former as an explicit cross-modal bridge resolved the alignment failure of Phase 1: training loss dropped substantially and continued to decrease, the first configuration to show genuine convergence.
4. BBLIP (final)	Success	Replacing the single linear projection with a Linear-GELU-Linear MLP, upgrading LoRA to LoRA+, and adding the Bengali normalizer (Section 4) together produced stable convergence across both training stages.

Table 9: Development trajectory from an initial direct vision-to-decoder connection to the final BBLIP design. Each row's observation is the empirical basis for a design decision retained in the final architecture.

variants failed or under-performed. On three merged Bengali captioning datasets, BBLIP reaches competitive or superior BLEU-1/2 and METEOR relative to prior from-scratch Bengali captioning baselines, while a fully-trained Transformer baseline retains an edge on higher-order BLEU. We hope that both the final architecture and the documented failure modes are useful to future work adapting pretrained vision–language models to other low-resource languages.

References

[1] J. Li, D. Li, S. Savarese, and S. Hoi. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. *ICML*, 2023.

[2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.

[3] S. Hayou, N. Ghosh, and B. Yu. LoRA+: Efficient Low Rank Adaptation of Large Models. *ICML*, 2024. arXiv:2402.12354.

[4] A. Bhattacharjee, T. Hasan, W. U. Ahmad, K. Samin, M. S. Islam, A. Iqbal, M. S. Rahman, and R. Shahriyar. BanglaNLG and BanglaT5: Benchmarks and Resources for Evaluating Low-Resource Natural Language Generation in Bangla. *Findings of EACL*, 2023.

[5] G. Geigle, A. Jain, R. Timofte, and G. Glavaš. mBLIP: Efficient Bootstrapping of Multilingual Vision-LLMs. arXiv:2307.06930, 2023.

[6] Q. Ye, H. Xu, G. Xu, J. Ye, M. Yan, Y. Zhou, J. Wang, A. Hu, P. Shi, Y. Shi, C. Li, Y. Xu, H. Chen, J. Tian, Q. Qi, J. Zhang, and F. Huang. mPLUG-Owl: Modularization Empowers Large Language Models with Multimodality. arXiv:2304.14178, 2023.

[7] M. R. Bhuiyan et al. A Bengali image captioning system using a ResNet-50 encoder and a context-aware attentive BiGRU decoder. *Heliyon*, 2024.

[8] Bangla Image Caption Generation through CNN–Transformer Based Encoder–Decoder Network. 2021.

[9] CNN–CNN Merge captioning baseline, as reported on the BanglaLekha Captions dataset.

[10] Visual-Attention captioning baseline, as reported on the BanglaLekha Captions dataset.

[11] Transformer-based captioning baseline, as reported on the BanglaLekha Captions dataset.

[12] M. Hodosh, P. Young, and J. Hockenmaier. Framing Image Description as a Ranking Task: Data, Models and Evaluation Metrics. *JAIR*, 2013.

[13] A. Radford et al. Learning Transferable Visual Models From Natural Language Supervision. *ICML*, 2021.

[14] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale. *NeurIPS*, 2022.